

Malik Choset

US Citizen · 412-995-8642 · mchoset@andrew.cmu.edu · [linkedin.com/in/malik-choset](https://www.linkedin.com/in/malik-choset) · tinyurl.com/malik-choset-portfolio

EDUCATION

Carnegie Mellon University

B.S. in Mechanical Engineering | GPA: 4.00

Pittsburgh, PA

May 2029

EXPERIENCE

Allen Control Systems

Mechanical Engineering Intern

Austin, TX

May 2026 – Present

- Designed an end-of-line test rig for an autonomous counter-drone solution utilizing, ensuring all finalized units successfully met operational requirements prior to deployment
- Optimized the testing rig for mass production by implementing robust Design for Manufacturing and Assembly principles, establishing a scalable tool and process that minimizes workflow bottlenecks on the assembly line
- Created engineering drawings complying with ASME Y14.5 standards, facilitating the rapid and accurate fabrication of all components
- Performed tolerance stack-up analysis on critical mechanism interfaces to account for expected manufacturing variations, ensuring all final dimensions remained strictly within specification

Carnegie Mellon Racing

Rear Wing System Lead

Pittsburgh, PA

May 2026 – Present

- Designing an adjustable rear wing system in SolidWorks prioritizing development of variable element configurations, enabling rapid tuning to suit specific dynamic events
- Developed parameterized CFD models, automating the analysis pipeline and drastically reducing setup time per design iteration
- Evaluated hundreds of aerodynamic configurations by executing geometric sweeps and identifying the optimal configurations
- Optimizing system structural integrity by performing FEA simulations in Ansys Mechanical, minimizing mass while maintaining structural goals

Drag Reduction System Lead

Sept. 2025 – May 2026

- Designed the active aerodynamics element for Formula Student Electric race car utilizing SolidWorks and Ansys Fluent, achieving up to a 45% reduction in drag between open and closed positions
- Validated system actuation and performance by developing free body diagrams and performed computation in Matlab, confirming the mechanism successfully met all goals
- Optimized system structural integrity using Ansys Mechanical, reducing the overall mechanism mass by over 75% compared to the previous year's design while maintaining adequate structural performance

Robotics Institute

Modular Robotics Research Assistant

Pittsburgh, PA

June 2025 – August 2025

- Developed a compact 2 Degree of Freedom module for the EigenBot system by optimizing motor packaging and material selections using Fusion 360, reducing the overall module form factor by 20 percent
- Developed a force-sensing gripper and foot module for the EigenBot modular robot system by engineering a compressible structure incorporating an internal magnet driven by FEA results from Ansys Mechanical, establishing novel force feedback capabilities to optimize locomotion and grasping

Fox Chapel Area School District

Summer Custodian

Pittsburgh, PA

June 2022 – August 2024

PROJECTS

Cycloidal Drive Generator · Python, SolidWorks API

April 2026

- Developed an interactive graphical app for parametric cycloidal drive generation utilizing Python to compute exact coordinate matrices and validate user-defined parameters, preventing invalid gear geometry while enabling rapid visual feedback and the immediate export gear drives
- Automated the 3D modeling of gear components by integrating with the SolidWorks API, eliminating CAD modeling time
- Automated a manufacturing pipeline by scripting the extraction from SolidWorks, creating 3D printable files

TECHNICAL SKILLS

Mechanical Design & Analysis: DFM, DFA, FEA, GD&T, Tolerance Stack-up Analysis

Design Software: SolidWorks, Ansys SpaceClaim, Ansys Discovery, Fusion 360

Analysis Software: Ansys Mechanical, Ansys Fluent, Luminary CFD

Manufacturing: CNC Mill, Manual Mill, Manual Lathe, 3D Printing

Data Analysis: Python, Matlab, Microsoft Excel